

Corrective Action Report —

CAR-2025-006

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Author	N. Halverson, EHS & Reliability Compliance Auditor; reviewed by D. Pham, Compliance Lead
CAR Status	Open
Subject	Fatigue Management Policy FMP-2 Repeatedly Exceeded at NW-PowerPool During Spring Vibration Cascade; Procedural Shortcuts Identified in TRA-10561 Servicing
Affected Region / Site / Asset	NW-PowerPool / NW-PowerPool-WIND-003, NW-PowerPool-WIND-014, NW-PowerPool-WIND-024 / TRA-10561, WIN-10058, WIN-10016, SUB-10641
Severity	High
Owner	M. Sutter, NW-PowerPool Plant Manager; co-owner H. Brody, Director EHS
Original Target Close Date	2025-11-30 (missed)
Revised Target Close Date	2026-03-31
Distribution	VP Operations; CHRO; Director EHS; Internal Audit; IBEW Local 125 (union steward, courtesy copy)

1. Finding

Fatigue Management Policy **FMP-2** ("Technician Overtime and Fatigue Risk Management"), revised 2024-08-12, limits scheduled and unscheduled overtime for technicians performing safety-critical work to **60 hours per rolling 7-day period**, with absolute limit of 14 consecutive workdays. Time-and-attendance records (system: Kronos-CR; period 2025-05-01 through 2025-06-30) confirm that at NW-PowerPool maintenance technicians logged weekly totals of **70 to 85 hours** for five consecutive

weeks during the Spring Vibration Cascade response, with two technicians exceeding 14 consecutive workdays.

This exceedance is correlated with procedural-shortcut findings during the servicing of TRA-10561 (transformer, NW-PowerPool-WIND-003) on 2025-05-26 (response to OUT-2025-1405). Specifically, the CSM-7 oil-spectrometer step (cross-reference CAR-2025-003) was attested as complete on the digital work order but no sample was logged at the testing lab. The technician on record had logged 78 hours in the trailing 7 days.

2. Evidence

2.1 Time-and-attendance exceedances (Kronos-CR audit)

Technician (ID-redacted)	Role	Week	Hours Logged	FMP-2 Limit	Ex- ceedance
Tech-NW-217	Wind drivetrain lead	2025-05-04 to 05-10	76.5	60	+27.5%
Tech-NW-217		2025-05-11 to 05-17	81.0	60	+35.0%
Tech-NW-217		2025-05-18 to 05-24	84.5	60	+40.8%
Tech-NW-217		2025-05-25 to 05-31	78.0	60	+30.0%
Tech-NW-217		2025-06-01 to 06-07	70.5	60	+17.5%
Tech-NW-409	Substation tech II	2025-05-11 to 05-17	79.5	60	+32.5%
Tech-NW-409		2025-05-18 to 05-24	82.5	60	+37.5%
Tech-NW-409		2025-05-25 to 05-31	76.0	60	+26.7%
Tech-NW-554	Wind tech I	2025-05-25 to 05-31	71.5	60	+19.2%
Tech-NW-554		2025-06-01 to 06-07	73.0	60	+21.7%

Tech-NW-217 worked **19 consecutive days** (2025-05-11 through 2025-05-29), exceeding the 14-day absolute limit. Tech-NW-409 worked **17 consecutive days**.

2.2 Procedural-shortcut findings during high-fatigue window

- **TRA-10561 service, 2025-05-26 (OUT-2025-1405 response):** CSM-7 §4.2 oil-spectrometer step attested complete in CMMS work order WO-2025-01405-EMR. No corresponding sample submittal exists in the Cascadia lab management system. Technician of record: Tech-NW-217 (weekly hours: 84.5). Cross-reference RCA-2025-002.
- **WIN-10058 service, 2025-05-04 (OUT-2025-0433 response):** Bolt-torque audit on yaw-bearing mounting (CSM-7 §3.4) attested complete in 11 minutes. Reference time per Procedure Time Standard PTS-2024-018 is 42–55 minutes for a 96-bolt yaw ring. Plausibility flag raised by Maintenance Engineering during quarterly audit.
- **SUB-10641 service, 2025-06-08:** Breaker-timing test performed without independent verification (FMP-2 §5.1 requires two-person verification for safety-critical tasks). Tech-NW-409 (weekly hours 82.5) signed both the performance and verification fields.
- **WIN-10016 service, 2025-06-21:** Drivetrain alignment recorded as within tolerance against a measurement that, on re-audit, falls outside MS-2025-03 §5.4 tolerance band.

2.3 Causation hypothesis

The Spring Vibration Cascade produced an emergent maintenance demand of approximately 1,840 labor-hours across May–June, against a planned regional capacity of approximately 1,200 hours. Apex was activated six times (cross-reference CAR-2025-004) but Apex labor does not substitute for Cascadia-employee technicians on internally-owned procedures and quality-attestation work. The remaining demand drove technician overtime without compensating fatigue-management controls.

3. Root Cause Hypothesis

- **Primary:** Local supervisory authority allowed FMP-2 exceedance without escalation. FMP-2 §6.3 requires CHRO and Director EHS notification at any individual exceedance of 70 hr/week; no such notification was filed in May or June 2025.
- **Contributing:** The exceedances were a structural response to the maintenance backlog created by the Q1 budget freeze deferrals (cross-reference

CAR-2025-001). The same finance instrument that suppressed PM execution then drove fatigue-inducing emergency response.

- **Contributing:** No quality-attestation challenge process exists at NW-PowerPool for work performed during declared high-fatigue conditions. Procedural shortcuts go undetected unless flagged by post-event audit.

4. Conflict With Existing Guidance

- **FMP-2 §3.1 and §6.3** (Cascadia internal) — explicitly violated.
- **OSHA 29 CFR 1910.269** — does not specify hour limits but does require fatigue-aware practice for high-voltage work. The two-person verification shortcut at SUB-10641 arguably implicates this regulation.
- **IBEW Local 125 collective bargaining agreement** Article 14 limits voluntary overtime to 16 hr/day. The 19-consecutive-day pattern is consistent with hour limits per day but exceeds the spirit of Article 14 on rest cycles.
- The exceedances arise in part from the operating posture established by **EPA-2024-11** (cross-reference CAR-2025-004), which assumes Apex absorbs emergency response while in fact Apex cannot perform attested-quality work on Cascadia-owned procedures.

5. Recommended Corrective Action

1. Restore FMP-2 compliance by establishing a CMMS-integrated weekly-hours gate: technicians at 55 hr/week receive a system warning; at 60 hr/week the system refuses new work-order assignment without site-supervisor + EHS dual approval.
2. Implement an attestation-quality challenge protocol for any closed work order where the technician's trailing 7-day hours exceed 60. Sample 25% of such work orders for documentation re-audit and physical re-inspection.
3. Backfill the four work orders cited in §2.2 with physical re-inspection of the cited steps (oil spectrometer, yaw-ring torque, breaker timing, drivetrain alignment).
4. Provide Director EHS standing authority to halt non-emergency work-order assignment to any technician at 65 hr/week.
5. Integrate fatigue-risk forecasting into emergency-response planning (e.g., when an outage is declared, system computes labor-pool fatigue posture and recommends Apex activation or internal response accordingly).

6. Action Owner & Timeline

Sub-action	Owner	Target Date
CMMS hours-gate deployment	R. Vasquez (CMMS) + H. Brody (EHS)	2025-12-15 — missed
Re-audit of 4 work orders	NW-PowerPool QA team	2025-09-30 — partially complete (3 of 4)
Attestation-quality challenge protocol	EHS + Reliability & Compliance	2026-02-28
EHS standing-halt authority memo	CHRO + Director EHS	2025-10-31 — missed
Fatigue-risk forecasting tool	CMMS team	2026-04-30

7. Implementation Status

Open. Three of four re-audits have been completed; the fourth (WIN-10016 alignment) was found out-of-tolerance and triggered a follow-on Engineering Review request. Tech-NW-217 was placed on mandatory leave 2025-06-30 through 2025-07-21 following EHS escalation; Tech-NW-409 received a written caution. No structural CMMS gate has been deployed.

Cross-references: FMP-2 §3.1, §6.3, §5.1 (internal policy basis); CAR-2025-001 (root upstream cause via budget-freeze backlog); CAR-2025-003 (CSM-7 oil-spectrometer step that was shortcut on TRA-10561); CAR-2025-004 (Apex contract structure that should have absorbed but did not); CAR-2025-009 (Engineering Review escalation policy); RCA-2025-002 (TRA-10561 root cause analysis); OIR-2025-003 (Period A outage investigation); IBEW Local 125 CBA Article 14; OSHA 29 CFR 1910.269.

End of report.